

entering into the cross-currency swap. If it does so, it will create synthetic floating rate funding of JPY LIBOR + 22 bps, a savings of 3 bps running compared to the direct issuance of a yen floater with a coupon of JPY LIBOR + 25 bps. Figure 2.9 shows the mechanics faced by the corporation if it issues in dollars and enters into the cross-currency swap. At the time the bond is issued, the corporation receives \$100mm from the bondholders, but it delivers this amount to the dealer in exchange for 8.53bn yen. Over the life of the transaction, the dealer pays USD LIBOR + 27 coupons to the corporation, which it uses to pay the bondholders. The corporation is left paying JPY LIBOR + 22. And at maturity, the dealer pays \$100mm principal to the corporation, which it uses to return the principal to the bondholders, and it pays the 8.53bn yen to the dealer.

As shown in this figure, by issuing in dollars and entering into the cross-currency swap, the corporation synthetically raises 8.53bn yen, funds itself at JPY LIBOR + 22, and returns 8.53bn yen at maturity. Notice that even though the margin on the dollar issuance was higher than on the yen-denominated issuance (27 bps versus 25 bps), it was nonetheless advantageous for the corporate to issue in dollars and swap due to the pricing inherent in the basis swap.

2.5.2 Asset Swaps

An *asset swap* is a popular trade in which an investor simultaneously purchases a bond and uses a swap to transform the nature of the asset from, say, a fixed one to a floating one.³⁶ Figure 2.10 shows how an investor who has purchased a fixed rate bond asset swaps the bond into a floater. There are a variety of different methodologies that are used to asset swap bonds. In the 1990s, a common type of asset swap transaction was known as a *par asset swap*. An investor would buy a fixed rate bond and enter into a swap in which he would pay fixed. The terms of the swap would be such that there would be an upfront payment made in the swap such that the *dirty price* of the bond, which is the quoted price plus accrued, plus the upfront amount would equal par, or the face amount of the bond. Thus, if the dirty price of the bond were less than par, the swap would include a payment from the investor to the dealer, and vice versa.³⁷ The terms of the swap would be such that the investor would pay fixed coupons in the swap in the exact amount and on the exact dates in which the bond payments were received.

³⁶Most asset swaps involve the purchase of a fixed rate bond, although some involve the purchase of a floating rate bond and use of a swap to transform the coupons to fixed.

³⁷Note that once the upfront payment has been made at the outset of the swap, it is no longer an NPV = 0 transaction. Suppose that in the terms of the asset swap the fixed

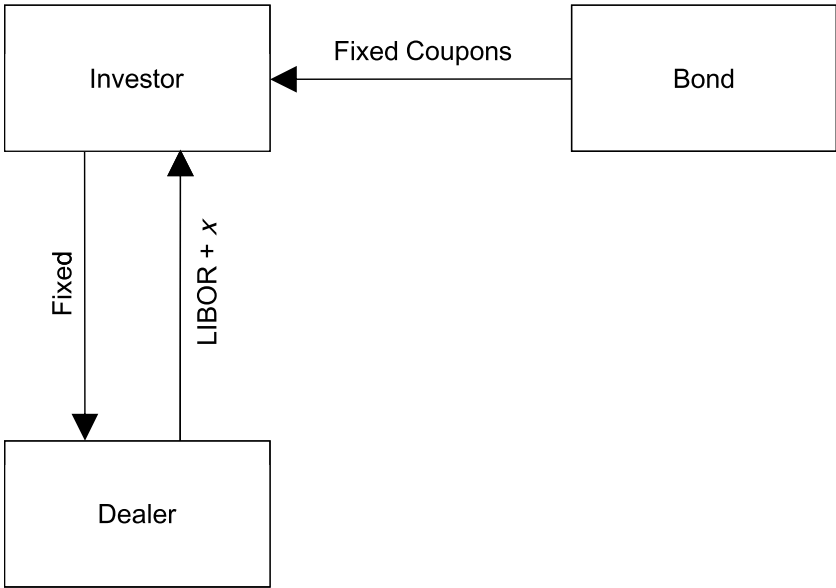


Figure 2.10: An Asset Swap

In return, the investor would receive LIBOR plus or minus a spread. This spread is known as the *asset swap spread*. An investor doing a par asset swap would want to receive as high of a spread to LIBOR as possible; a high spread would indicate that the bond was “cheap on a LIBOR basis,” and a low spread would indicate that the bond was “rich on a LIBOR basis.”

More recently, the variety of asset swap most often employed results in a much simpler transaction. It is now more common for an investor to buy the bond and pay fixed in a swap that has a maturity date equal to the maturity of the bond. In this type of transaction there is no upfront payment, and the fixed cash flows in the swap do not in general exactly match the coupons in the bond. The difference between the swap rate that matures when the bond matures and the yield of the bond is referred to as the *yield/yield asset swap spread*. As in the case of a par asset swap, the higher the asset swap spread, the cheaper the bond is considered, and vice versa.

We’ll have more to say about asset swaps in Chapter 8. For now, here are a few takeaways to keep in mind. First, when an investor buys a fixed

rate payer makes an upfront payment of \$100,000 to its counterparty. Then after that upfront payment in the swap has been made, the value of the remaining cash flows in the swap is \$100,000 in favor of the fixed rate payer.

rate bond and asset swaps it, he no longer has material duration exposure on the bond since he is transforming it to a floater. That is, the package of the bond and the swap together have no (or little) sensitivity to movements in interest rates. Second, the investor still has exposure to the underlying credit he is buying; if the credit spread on the bond should tighten, he will benefit, and vice versa. Third (and this should not necessarily be clear at this point in time), whereas an investor who buys the bond and asset swaps it is going to benefit if the credit spread of the bond tightens, he will also benefit if swaps spreads in the offsetting swap widen. Again, we will look at these dynamics much more closely in Chapter 8.

2.5.3 Balance Sheet Management

In Section 2.2.1, we alluded to the fact that swaps are used for balance sheet management purposes. Remember how swap spreads widened dramatically in the summer of 2003 because of all of those mortgage portfolios who were paying fixed? As the market sold off, mortgage securities were extending, which increased the duration of mortgage portfolios. Due to the negative convexity of mortgages, mortgage portfolios were getting longer and longer

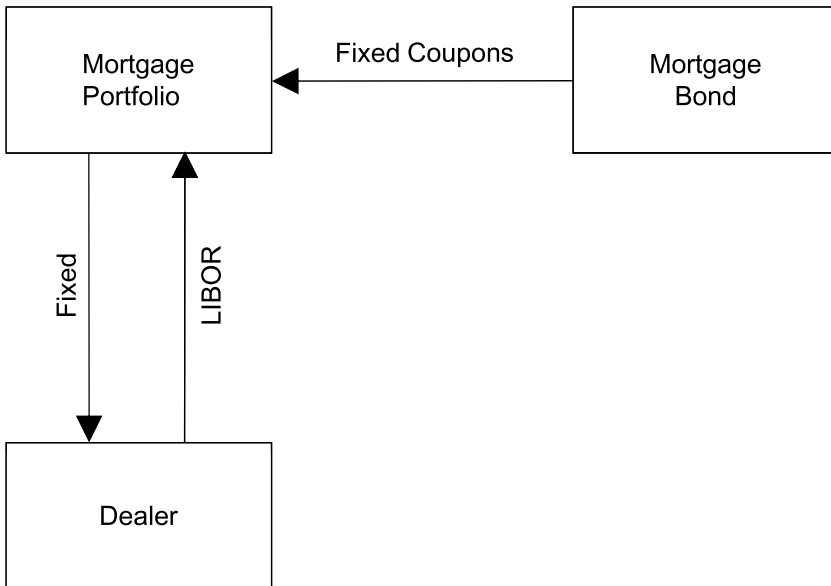


Figure 2.11: Mortgage Portfolio Paying Fixed to Manage Duration Risk

the market as rates rose. This is of course a bad thing for those portfolios, because as the market sells off, the last thing you want is to essentially have more fixed rate bonds in your portfolio than you started with, which are decreasing in value as rates rise. But this is the plight of those whose portfolios exhibit negative convexity. And during the summer of 2003, many mortgage portfolios managed this increase in the duration of their portfolios by paying fixed in swaps. This dynamic is depicted in Figure 2.11.

Problem Set 2

1. A U.S. corporate would like to raise \$500mm by issuing debt. The corporation has determined that it would like to issue debt for a term of five years and would like to fund itself on a floating rate basis. The corporate could issue a floating rate bond, or it could issue a fixed rate bond and enter into a fixed-floating swap with a swap dealer to transform the nature of the coupons it is paying from fixed ones to floating ones, thereby creating what is known as “synthetic floating rate funding.” Assume that the corporation can issue 5-year fixed paper at par with a coupon of $T_5 + 96$ bps, where T_5 represents the yield on the on-the-run 5-year Treasury at the time of pricing, or that it can issue a 5-year floater at 3-month LIBOR + 66 bps.

Suppose that the market is as follows:

Maturity	Treasury Yield	Swap Spread
Year 1	5.00%	26
Year 2	5.11%	28
Year 3	5.30%	32
Year 4	5.47%	33
Year 5	5.60%	34
Year 7	5.70%	36
Year 10	5.90%	39
Year 30	6.20%	42

In order to price a transaction of this size, the swap dealer agrees to price the swap using mid-market Treasury yields (as compared to using the relevant bid or offer side Treasury yield) but will price swap spreads 2 bps from the mid-market swap spread levels shown in the table above. Based on these numbers, would you suggest that the corporate issue a floating rate bond or that the corporate issue a fixed rate bond and swap to floating? Explain.

2. Suppose you are an investment banker and one of your clients is a U.S. corporate that would like to raise \$500mm by issuing debt in order to fund a domestic project. The corporation has determined that it would like issue debt for a term of five years. Your capital markets group has determined that domestic investors would have interest in buying a floater issued by this corporate. The corporation is pleased to hear this, as it desires to fund itself in dollars on a floating rate basis. Given the work of your capital mar-

kets group, it appears that investor interest for a 5-year floater issued by the corporate is at 3-month USD LIBOR + 10 bps.

As an alternative, your capital markets group also indicates that Japanese investors would have interest in buying a yen-denominated floater issued by this U.S. corporate. In particular, Japanese investor interest for a 5-year floater issued by the U.S. corporate in yen is at 3-month JPY LIBOR + 5 bps. Assume the current \$/yen exchange rate is 109, and thus the corporation would want to issue bonds with a face amount of 54.5bn yen in order to raise an equivalent amount of cash. Of course, if the U.S. corporate were to issue in yen, it would want to convert the proceeds of the issuance into dollars since it is looking to raise the money to fund a domestic project. Furthermore, assume that if the corporate were to issue in yen, it would want to convert its floating yen liability for the next five years into floating dollars.

You check with your swap desk to obtain pricing on a dollar/yen floating-floating cross-currency swap (including principal exchanges upfront and at maturity) with a maturity of five years. You are told that if one counterparty pays coupons of 3-month USD LIBOR, then the other counterparty would have to pay 3-month JPY LIBOR + 6 bps. This is a mid-market quote and does not incorporate any bid/offer for the dealer. Equivalently, one counterparty would pay 3-month USD LIBOR – 6 bps, and the other counterparty would pay 3-month JPY LIBOR at mid-market on this cross-currency swap.³⁸ Furthermore, your swap desk indicates that it would be willing to transact either side of this cross-currency swap 3 bps from mid.

Given this information, what would you advise your client to do? Explain.

3. Suppose that the swap curve is as in Table 2.1, with PV01s as given in Table 2.3. Base your answers to the following questions on this swap curve and these PV01 numbers. Assume that all transactions are done at mid-market.

³⁸Technically, this is not entirely accurate. We are not quite free to “add or subtract” basis points at will from one side of the swap to the other, despite the fact that payments on both sides of this swap are made on a quarterly Act/360 basis. To be precise, we would need to make an adjustment to the spread when moving bps from one side of the swap to the other. The reason for this is that the yen side of the swap is of course priced off of the yen swap curve, and the dollar side is priced off of the dollar swap curve. Since yen rates differ from dollar rates, the PV01 of a bp on one side does not equal the PV01 on the other side. Assume we can totally ignore this fact for this problem, and thus we can feel free to move basis points from one side of the swap to the other without consideration of this fact.

a) Suppose that a swap dealer and a client are entering into \$50mm of a 5-year swap in which the dealer is paying fixed. The dealer inadvertently quotes a fixed rate that is 5 bps above the market, and the trade is executed as such. How much is this “mistake” worth and in whose favor is it in?

b) Suppose that a swap dealer and the counterparty are entering into \$50mm of a 5-year swap in which the dealer is paying fixed. After entering into the trade, assume that the swap curve rallies 5 bps. How much is the swap worth (i.e., what is the current mark-to-market of the swap) and in whose favor is it in?

c) Suppose that a swap dealer and a counterparty are entering into \$100mm of a 5-year swap in which the dealer is paying fixed. The terms of the trade are a bit nonstandard, however. The counterparty would like to receive an upfront payment of \$1mm at the outset of the trade. The fixed rate in the trade should be adjusted to account for this upfront payment. What fixed rate should the dealer pay?

d) Suppose that the swap dealer and the counterparty are entering into \$100mm of a 30-year swap in which the dealer is paying fixed. Like part c), the terms of the trade are a bit nonstandard, however. The counterparty would like to receive an upfront payment of \$1mm at the outset of the trade. The fixed rate in the trade should be adjusted to account for this upfront payment. What fixed rate should the dealer pay?

e) How does your answer to part c) compare to your answer to part d)? Explain.

f) Consider a trade that is very similar to part d), except that the counterparty would like to receive a payment of \$1mm at the maturity of the trade in 30 years, instead of at the effective date of the trade. The fixed rate in the 30-year swap will be adjusted to reflect this payment at maturity. How does the fixed coupon in this swap compare to the fixed coupon you solved for in part d)? In other words, is it higher, lower, or the same as the coupon you solved for in part d)?

g) Suppose that you are the chief risk officer who works at the swap dealer. Your job is to try to minimize counterparty credit exposure whenever possible. Given the choice, do you prefer that the swap dealer enter into the trade in part d) or the trade in part f)? Explain.

h) Suppose that the swap dealer pays fixed on \$100mm 2-year swaps to a client and receives fixed on \$100mm 5-year swaps from the client. Both trades are executed at market rates. Immediately after the trades are executed, the client requests that the terms of the trades are adjusted. In particular, the fixed rate on the 5-year swap should be increased by 1 bp. The fixed rate on the 2-year swap should be changed so that it is not necessary for the counterparties to exchange an upfront fee. What is the new fixed rate in the 2-year swap?

i) This trade is identical to part h) with the notional on the 5-year swap \$100mm as above; however, now the notional on the 2-year swap is \$229.72mm. Given that the fixed rate on the 5-year swap is increased by 1 bp, how should the fixed rate on the 2-year swap be changed so that it is not necessary for the counterparties to exchange an upfront fee?

j) How does your answer to part h) compare to your answer to part i)? Explain.

k) Suppose that earlier today a client received on \$1bn 10-year swaps at a rate of 5.95%. The swap curve prevailing now is the one in Table 2.1. What is the current mark-to-market of the trade? That is, what is the value of the trade in the current swap curve environment?

4. Suppose that the swap curve is as in Table 2.1, with PV01s as given in Table 2.3. Suppose further that the 2-year, 3-year, 5-year, 7-year, 10-year, and 30-year on-the-run Treasuries are all trading currently with a dirty price of par. Suppose that it is widely anticipated in the market that each of these Treasuries will be going special for the next 30 days and, as such, the repo rate on each of these securities is anticipated to decrease by 100 bps for the next 30 days. How do you expect that this anticipated change in repo rates will affect the 2-year, 3-year, 5-year, 7-year, 10-year, and 30-year swap spread, respectively?

Chapter 3

The Pricing of Swaps

We spent a considerable amount of time in Chapter 1 discussing the fundamentals of the swaps market. In Chapter 2 we focused on the risk characteristics of swaps. In this chapter we are going to be focused on the pricing of swaps.

When we speak of pricing a fixed-floating swap when it is first executed, this amounts to determining the fixed coupon that makes the trade an $NPV=0$ transaction. At other times we will be interested in valuing a fixed-floating swap that has been in existence for some time. In these cases, the pricing of a swap refers to determining the mark-to-market value of a swap whose fixed coupon was previously set. Prior to addressing both of these questions, we will examine the relationship between swap rates and Euro-dollar futures.

3.1 Where Do Swap Rates Come From?

There is an intimate relationship between swap rates and Eurodollar futures contracts.¹ In order to understand this relationship, we will first go through the very basics of Eurodollar futures. Eurodollars are time deposits made

¹This is true for swaps with maturities of five years and less. Beyond five years, Euro-dollar futures contracts are not considered very liquid by many traders, and as a result they are typically not used to price swaps with a maturity of greater than five years. We will implicitly restrict our attention, then, to the pricing of swaps on the short end of the curve in this section.

in U.S. dollars but are actually held at banks outside of the United States. The rates of interest implicit in these deposits have long been a standard benchmark in the market. A Eurodollar futures contract is traded on the Chicago Mercantile Exchange and²

- Is a commitment to deliver a \$1mm Eurodollar time deposit with a 3-month maturity;
- Has a price that is quoted as an index value equal to 100 minus the Eurodollar interest rate (thus, e.g., a rate of 3% has a price quote of 97.00);
- Has contracts months that are March, June, September, and December for a total of 10 years (40 contracts);³
- Has as its last trading day two London business days before the third Wednesday of the contract month;
- Has a cash settlement price equal to 100 minus the BBA survey of 3-month USD LIBOR on the last trading day;
- Has \$25 in value for a 1 bp move in the underlying rate (a “tick”); contracts may trade in half-tick increments, with the exception of the nearest-term contract, which can trade in quarter-tick increments.

Although Eurodollar futures are indeed futures and thus have daily variation margin, we will for our purposes here ignore this feature and regard them as forwards.⁴

²For more detail on Eurodollar futures, see Burghardt (2003) or Hull (2008), for example; or for information from the Chicago Mercantile Exchange, go to <http://www.cmegroup.com/trading/interest-rates/stir/eurodollar.html>.

³In addition, the four nearest serial months that are not of this cycle also trade.

⁴A forward contract is an agreement to buy or sell some underlying asset at a specified time (which we will refer to as the *settlement date*) at a price that is specified today. That price that is specified today is known as the *forward price*. No cash exchanges hands today, and in fact the only time cash changes hands is at the settlement of the contract. A futures contract is similar to a forward contract, except there is a daily marking-to-market of the contract. Counterparties in a futures contract are subject to daily variation margin, reflecting changes in the value of the contract. Forwards and futures are very similar except that futures have cash flows that occur over the life of the contract. The price at which the underlying asset will be bought or sold in a futures contract is referred to as the *futures price*.

Are forward prices equal to futures prices? This is a question that has been analyzed extensively. Cox, Ingersoll, and Ross (1981) and Jarrow and Oldfield (1981) both determined that futures prices and forward prices can differ under the assumption that interest rates are stochastic. If interest rates are stochastic, then futures prices will be greater than forward prices, provided that futures prices are positively correlated with interest